

Discrete Random Variables

Families of random variables

In the sequel, we will list the distribution series of some of the most useful discrete random variables. In you are using the electronic version of this file, you can click on their name for further information.

► **binomial random variable** $X \sim \text{Bin}(n, p)$ the random variable which counts the number of successes in n trials of a binomial experiment

$$X : \begin{pmatrix} 0 & 1 & \dots & k & \dots & n \\ q^n & C_n^1 \cdot p \cdot q^{n-1} & \dots & C_n^k \cdot p^k \cdot q^{n-k} & \dots & p^n \end{pmatrix}$$

where p and $q = 1 - p$ are the probabilities of a success, respectively failure, at each independent trial

• the mean and the variance are

$$E(X) = n \cdot p \quad \text{var}(X) = n \cdot p \cdot (1 - p)$$

► **geometric random variable** $X \sim \text{Geo}(p)$ is the appropriate model when, in a binomial experiment, we count the number of failures until the first success occurs

$$X : \begin{pmatrix} 0 & 1 & \dots & k & \dots \\ p & p(1-p) & \dots & p(1-p)^k & \dots \end{pmatrix}$$

• the mean and the variance are

$$E(X) = \frac{1-p}{p} \quad \text{var}(X) = \frac{1-p}{p^2}$$

► **Bernoulli's random variable** $X \sim \text{Ber}(p)$ it can be thought of as a model for the set of possible outcomes of any single experiment that asks a yes-no question

$$X : \begin{pmatrix} 0 & 1 \\ 1-p & p \end{pmatrix}$$

• the mean and the variance are

$$E(X) = p \quad \text{var}(X) = p(1-p)$$

► **uniform random variable** $X \sim \mathcal{U}(k, \ell)$ is a probability distribution wherein a finite number of values are equally likely to be observed

$$X : \begin{pmatrix} k & k+1 & \dots & \ell \\ \frac{1}{\ell-k+1} & \frac{1}{\ell-k+1} & \dots & \frac{1}{\ell-k+1} \end{pmatrix}$$

• the mean and the variance are

$$E(X) = \frac{k+\ell}{2} \quad \text{var}(X) = \frac{(\ell-k+1)^2 - 1}{12}$$

► **hypergeometric random variable** $X \sim \text{Hyp}(N, K, n)$: describes the probability of k successes (random draws for which the object drawn has a specified feature) in n draws, without replacement, from a finite population of size N that contains exactly K objects with that feature, wherein each draw is either a success or a failure

• its probability mass function is $P(X = k) = \frac{C_K^k \cdot C_{N-K}^{n-k}}{C_N^n}$ if $\max\{0, n + K - N\} \leq k \leq \min\{n, K\}$ and 0 otherwise

• the mean and the variance are

$$E(X) = n \cdot \frac{K}{N} \quad \text{var}(X) = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1}$$

► **negative binomial random variable** $X \sim \text{NB}(r, p)$: is a probability distribution of the number of successes, in a binomial experiment, before a specified number r of failures occurs.

$$X : \begin{pmatrix} 0 & 1 & \dots & k & \dots \\ (1-p)^r & C_{1+r-1}^1 \cdot p \cdot (1-p)^r & \dots & C_{k+r-1}^k \cdot p^k \cdot (1-p)^r & \dots \end{pmatrix}$$

• the mean and the variance are

$$E(X) = r \cdot \frac{1-p}{p} \quad \text{var}(X) = r \cdot \frac{1-p}{p^2}$$

• sometimes we want to count the number of trials needed to produce the r -th success, such a random variable X will have the probability mass function

$$P(X = k) = C_{k-1}^{r-1} \cdot p^r \cdot (1-p)^{k-r} \quad k = r, r+1, r+2, \dots$$

► **Poisson's random variable** $X \sim \text{Po}(\lambda)$: a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event

$$X : \begin{pmatrix} 0 & 1 & \dots & k & \dots \\ e^{-\lambda} & \frac{\lambda}{1!} \cdot e^{-\lambda} & \dots & \frac{\lambda^k}{k!} \cdot e^{-\lambda} & \dots \end{pmatrix}$$

• the mean and the variance are $E(X) = \lambda$ and $\text{var}(X) = \lambda$

Random variable

A random variable is a mathematical model for a random experiment. When the possible outcomes of the experiment are countable the resulted model is a *discrete* random variable. A discrete random variable will *count* something, a continuous random variable will *measure* something.

• in general a discrete random variable is given by its **distribution series**

$$X : \begin{pmatrix} x_i \\ p_i \end{pmatrix}_{i \in I}$$

where x_i , called the *values* of X , are the possible outcomes of the experiment and $p_i = P(X = x_i)$ the probability of each outcome

• the **probability mass function (PMF)** of the discrete random variable X is

$$P_X(x) = P(X = x)$$

the function that attains to x the probability of the event $X = x$

• the **cumulative distribution function (CDF)** of a random variable X is

$$F_X(x) = P(X \leq x)$$

Mean and variance

► the following identities hold

$$E(X + aY) = E(X) + a \cdot E(Y) \\ \text{var}(aX + C) = a^2 \cdot \text{var}(X)$$

where C is the constant random variable

• if X and Y are **independent**

$$E(X \cdot Y) = E(X) \cdot E(Y) \\ \text{var}(X + aY) = \text{var}(X) + a^2 \cdot \text{var}(Y).$$

Functions of random variables

► for a discrete random variable X the PMF of the function $Y = g(X)$ is

$$P_Y(y) = \sum_{x: g(x)=y} P_X(x)$$

and the expected value

$$E[Y] = \sum_{x \in S_X} g(x) \cdot P_X(x)$$

Conditional probability models

► given an event B , with $P(B) > 0$ the conditional probability mass function of X is

$$P_{X|B}(x) = \begin{cases} \frac{P_X(x)}{P(B)} & , \text{ if } x \in B \\ 0 & , \text{ otherwise} \end{cases}$$

• the conditional expected value

$$E[X|B] = \sum_{x \in B} x \cdot P_{X|B}(x)$$

• the conditional variance is

$$\text{var}(X|B) = E[X^2|B] - E^2[X|B]$$

► a random variable X resulting from an experiment with event space $B_1, B_2, \dots, B_m$ has the PMF

$$P_X(x) = \sum_{i=1}^m P_{X|B_i}(x) \cdot P(B_i)$$

and similarly the expected value

$$E(X) = \sum_{i=1}^m E[X|B_i] \cdot P(B_i)$$